

Analysing the impact of renewable electricity support schemes on power prices: The case of wind electricity in Spain

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ABSTRACT

It is sometimes argued that renewables are “expensive”. However, although it is generally true that the private costs of renewable electricity generation are certainly above those of conventional electricity, that statement fails to consider the social benefits provided by electricity from renewable energy sources (RES-E), including environmental and socioeconomic ones. This paper empirically analyses an additional albeit usually neglected benefit: the reduction in the wholesale price of electricity as a result of more RES-E generation being fed into the grid. The case of wind generation in Spain shows that this reduction is greater than the increase in the costs for the consumers arising from the RES-E support scheme (the feed-in tariffs), which are charged to the final consumer. Therefore, a net reduction in the retail electricity price results, which is positive from a consumer point of view. This provides an additional argument for RES-E support and contradicts one of the usual arguments against RES-E deployment: the excessive burden on the consumer.

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1. Introduction

Widespread public support for electricity from renewable energy sources (RES-E) has been justified according to environmental and socioeconomic reasons. On the one hand, RES-E pollutes much less than conventional (fossil-fuel-based) electricity and, therefore, avoids its negative environmental externalities. More specifically, RES-E contributes to the achievement of the climate change mitigation goals of the Kyoto Protocol (KP), while emissions of local pollutants are reduced as well. Apart from environmental benefits, RES-E provides several socioeconomic advantages, including diversification of energy supply, enhanced regional and rural development opportunities, creation of a domestic industry and employment opportunities.

Those benefits translate into a generally lower social cost (inclusive of private costs plus negative external costs minus positive externalities) for renewable energy, but market operators

decide on the basis of private and not social costs unless, of course, policy measures internalise those externalities. Public support to renewable electricity levels the playing field with respect to conventional electricity, internalises the positive externalities of renewable energy in the decisions taken by economic actors and allows renewable energy to penetrate the electricity market (del Río and Gual, 2004).

Generally, support for renewable electricity is finally paid by electricity consumers in their electricity bills (retail price). Especially in countries experiencing a significant increase of RES-E generation, such as Spain, there is a concern that this support places a significant and increasing financial burden on the final consumer. Consumer costs are the result of adding up the wholesale electricity price to RES-E support. If, as with feed-in tariffs (FITs), support is paid per kWh of RES-E generation to producers, then an increase in RES-E generation would translate into a greater amount of total payment and, thus, a greater financial burden for the consumer.

However, the impact of an increasing amount of RES-E on consumer costs may be more ambiguous than is often considered. To our knowledge, Jensen and Skytte (2003) were the first to point out that a greater RES-E deployment could even reduce final electricity prices because RES-E promotion encourages RES-E generation, which has lower variable costs than fossil-fuel conventional electricity. If fossil-fuel-fired plants, which are usually the marginal generation plant and the ones setting the wholesale electricity price, substitute for RES-E, then the

Abbreviations: EEX, European Energy Exchange; ETS, emissions trading schemes; FIT, feed-in tariffs; GWh, gigawatts/hour; KP, Kyoto Protocol; kWh, kilowatts/hour; MC, marginal costs; MWh, megawatts/hour; NSE, Non-supplied energy; OMEL, Spanish market operator; PV, photovoltaics; REE, Red Eléctrica de España (system operator); RES, renewable energy sources; RES-E, electricity from RES; RPS, renewable portfolio standard; TEP, tradable emissions permits; TGC, tradable green certificates.

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wholesale electricity price would be reduced. This reduction could offset the increase in final electricity prices resulting from RES-E support, leading to a net reduction in retail prices. They argue that, if this offset occurs, then it could be interesting to combine the RES-E promotion scheme with an emissions trading scheme (ETS) to achieve the three goals of RES-E deployment, cost-effective emissions reductions and low electricity prices.¹

Therefore, RES-E promotion could lead to a win–win situation in this context: an increase in RES-E deployment (and its associated environmental and socioeconomic benefits) and a reduction in electricity prices. Indeed, this is one of the factors behind the recent approval (4 August 2007) in the US Congress of a Federal Renewable Energy Standard, which, if finally approved by the Senate, would require utilities to generate 15% of their electricity from renewables by 2020, arguing that the extra expense of RES-E would be offset by a lower demand, and thus lower prices, for natural gas and coal (the marginal generation), leading to a reduction of electricity prices (*The Economist*, 2007).²

Furthermore, an additional impact, which was not considered by Jensen and Skytte (2003) should be taken into account: RES-E generation reduces the number of CO₂ allowances needed by the electricity generation sector to cover its emissions, if this sector is included in an ETS. This lower demand for allowances reduces their price and, thus, compliance costs, putting an additional downward pressure on the electricity price.

This paper aims to empirically analyse this issue with a case study of Spanish wind generation. This case is worldwide relevant, since Spain only ranks behind Germany regarding wind-installed capacity in the world and wind generation already covers 10% of electricity demand. It is shown that the reduction in the wholesale electricity price is greater than the additional cost increase arising from RES-E support (FITs), which are charged to final consumers. Therefore, the net effect on the retail electricity price would be positive from a consumer point of view (i.e., a reduction is experienced), providing an additional argument for RES-E support.

Accordingly, the paper is structured as follows. Section 2 provides a brief overview of the existing literature. Section 3 provides the analytical framework for this paper. Section 4 briefly discusses the situation and trends of wind electricity and its promotion in Spain. The empirical study is provided in Section 5. The paper closes with some concluding remarks and policy recommendations in Section 6.

2. The theoretical and empirical literature on interactions between RES-E deployment, CO₂ emissions reductions and the electricity market

The EU has three major goals in the environmental-energy realms: reduction of CO₂ emissions, increase in RES-E deployment and low electricity prices for consumers.³ In order to achieve

these (sometimes conflicting) goals, two instruments are available: RES-E support measures and ETS. These instruments coexist in the EU, with national RES-E promotion and an EU-wide ETS.

There are two major streams in the literature on the interactions between RES-E support schemes, ETS and the electricity market. However, this literature has been mainly theoretical and has focused mostly on the interaction between RES-E support schemes and emissions trading, whereas the empirical literature on the interactions between RES-E promotion and the electricity market is surprisingly thin (see del Río, 2007).⁴

2.1. The theoretical literature

One stream of the literature focuses on the theoretical interactions resulting from the simultaneous application of both instruments (RES-E promotion and ETS) at a theoretical level (see, among others, Jensen and Skytte, 2003; Skytte, 2006; Morthorst, 2001, 2003). The RES-E promotion instrument considered in these interactions has been a tradable green certificate scheme (TGC),⁵ although most conclusions could be extrapolated to the case of FITs.

At a national level, the interactions between RES-E deployment, ETS and the electricity market can be summarised as follows (Fig. 1):

- (1) *Electricity market and ETS.* The introduction of an ETS increases final electricity prices, since conventional electricity generators and/or electricity distributors tend to pass the extra costs of CO₂ emissions control to electricity prices.
- (2) *Electricity market and RES-E deployment.* An increase in RES-E deployment leads to ambiguous effects on consumers. On the one hand, the introduction of RES-E promotion schemes tends to reduce the electricity price, since they displace and reduce the demand for conventional (fossil-fuel-based) electricity (in so far as marginal generation is based on fossil fuels). On the other hand, the costs of RES-E support are passed to consumers.
- (3) *ETS and RES-E deployment.* The interaction between ETS and RES-E deployment/promotion is a complex one. The ETS makes RES-E deployment more competitive, since it increases the costs of conventional electricity. In turn, RES-E deployment reduces the demand for tradable emissions permits in an ETS, and, thus, their price.

Jensen and Skytte (2003) and Skytte (2006) analyse how to attain the three goals (RES-E deployment, CO₂ emissions reductions and low electricity prices) with two instruments (a RES-E promotion scheme and an ETS) and show that the choice of RES-E deployment and CO₂ emissions reduction targets (i.e., “quotas”) depends on the correlation between the final electricity price and the quotas. If RES-E deployment leads to a reduction in final electricity prices (negative correlation), then a RES-E quota should be applied, leading to a reduction in CO₂ emissions at (moderate) costs for the consumers. However, if RES-E deployment as a result of RES-E support increases final electricity prices, then the RES-E quota and the CO₂ quota should be combined in order to increase RES-E deployment. Therefore, with negative correlation it is better to choose a RES-E quota and no emissions quota. In the case of positive correlation, it is optimal to choose a combination of the

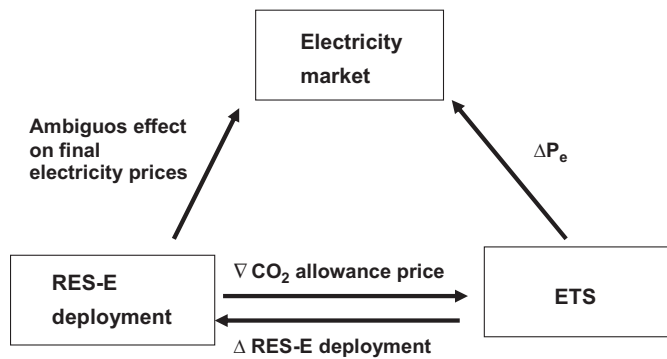
¹ For example, this benefit of RES-E generation is not considered by the European Commission itself when calculating the costs of increasing the share of RES to 20% of energy consumption in 2020 (European Commission, 2007). The Commission concludes that, if the savings of oil imports (with a price of oil reaching 78\$/barrel) and the reduction in CO₂ emissions are taken into account, the costs of reaching the renewable energy target would be negligible. If the costs of reduction in the retail price of electricity were considered, the conclusion could well be that the RES-E policy completely pays itself.

² One of these reports is from the consulting firm Wood Mackenzie (2006), which shows that the increase in RES-E necessary to reach the 15% target translates into an incremental construction cost of \$134 billion (2006 dollars) between 2006 and 2026, whereas the Federal RPS would lead to savings of \$240 billion (2006 dollars) in wholesale power costs, outweighing the higher capital investment to build such additional capacity.

³ The goal of low electricity prices (which is supposed to be one of the benefits of the electricity liberalisation process) has been stressed in several recent documents from the European Commission (see, among others, European Commission, 2006).

⁴ Only a summary of this literature is provided here. For a detailed overview, see del Río (2007).

⁵ See Morthorst (2001) for further details on the functioning of this instrument.



Source: Own elaboration.

Fig. 1. Summary of interactions between different markets/schemes. Source: Own elaboration.

RES-E and the emission quota. They found that it will never be optimal to regulate only through the emission quota no matter what the correlation is. Thus, the two instruments are to some extent substitutes for each other, i.e., there is a positive synergy in their use. Table 1 summarises this discussion. In spite of the relevance of their theoretical findings and their conclusion that the impact is ambiguous, Jensen and Skytte (2003) do not empirically test those relationships.

Therefore, taking into account the possible relationship between RES-E generation, the wholesale electricity price and the retail electricity price, three situations may exist:

- **Absolute negative correlation.** The reduction in the wholesale price of electricity more than compensates the increase in costs related to RES-E support, leading to a reduction in the retail electricity price and leaving the consumer better off than in the absence of RES-E support.
- **Relative negative correlation.** The reduction in the wholesale price of electricity does not compensate the increase in costs related to RES-E support, leading to an increase in the retail electricity price.
- **Positive correlation.** RES-E deployment does not lead to a reduction in the wholesale price of electricity. Therefore, the retail electricity price increases by the increase of RES-E support.⁶ In so far as we have assumed perfect competition in the electricity market and marginal cost pricing, positive correlation in the sense we have mentioned here is highly unlikely, at least in the short term. Given the existence of a certain amount of high marginal cost thermal generation and that wind generation has very low marginal costs, it is likely that an increase in RES-E generation (wind) will lead to a reduction in wholesale electricity prices.

In this context the price in the wholesale electricity market is the result of the intersection of the electricity demand curve and the supply curve (also called the merit order) and the plant with the highest marginal cost (usually a fossil-fuelled one) sets the wholesale price. An increase in low marginal cost RES-E generation would drive the currently marginal power plants (usually thermal plants) in the merit order out of the market. The increase in RES-E generation shifts the supply curve to the right, which would reduce the wholesale price of electricity.

This shift in the curve would not lead to a reduction in the wholesale price if the demand curve was totally elastic, which

Table 1
Optimisation criteria concerning final electricity prices

	Positive correlation ^a	Negative correlation ^a
Reaching the RES-E target	RES-E quota	RES-E quota
Reaching the emissions quota	Emissions quota	RES-E quota
Reaching both goals simultaneously	Combination of emissions and RES-E quota	RES-E quota

Source: Own elaboration.

^a Positive correlation indicates that an increase in the RES-E quota increases consumer (retail) prices. Negative correlation indicates the opposite.

would be highly unrealistic (actually the reverse is usually true, i.e., inelastic demand, as assumed in this exercise).

If the slope of the supply curve was very steep or if the share of RES-E was very high, a reduction in the wholesale price would be more likely than in the cases of a low increase in RES-E generation and a less steep supply curve. The latter would occur if the existing thermal-based generation showed very high marginal costs, i.e., if the marginal cost of existing thermal-base technologies increased very much with increasing generation. The greater the share of high marginal cost thermal generation, the greater the possibility that wind generation substitutes for thermal generation and the greater the possibility that wholesale power prices are reduced. In contrast, the aforementioned potential impact would be lower if the electricity mix had an enormous share of base-load capacity with low marginal costs (hydro and nuclear).

On the other hand, as empirically found by Unger and Ahlgren (2005), there is likely to be a threshold of the share of RES-E in total electricity below which a small increase in RES-E is unlikely to result in an increase in wholesale prices. In other words, the lower the increase in RES-E deployment and the lower the share of RES-E in overall electricity production, the lower the probability that there will an impact of an increase in RES-E generation on wholesale electricity prices. This would be irrespective of the market structure we assumed.

2.2. The empirical literature

One stream of the empirical literature uses case studies and another uses model simulations to analyse different aspects of the interaction between RES-E support schemes, ETS and the electricity market. A third one specifically focuses on the impact of RES-E promotion on electricity prices.

- The possible interactions in several countries has been analysed, using case studies (see Sorrell, 2003; Sijm, 2003, among others). These studies do not consider the simultaneous interaction between instruments in order to reach a target (Skytte, 2006), focus on the interaction between the EU ETS and RES-E support schemes at national level and no attention is paid to the impact of RES-E generation on the electricity market.
- Another stream of the empirical literature uses model simulations. For example, Hindsberger et al. (2003) apply the Balmore simulation model to analyse the interaction between TEPs and TGCs for the electricity sector in several Baltic countries. The TEP market comprises the four Nordic countries and the TGC market extends to Denmark, Finland, Sweden and Germany. The authors combine different levels of ambition for TGC and TEP quotas and find out that adding

⁶ Note that positive correlation for Jensen and Skytte (2003) differs from our use of the term. Positive correlation à la Jensen and Skytte encompasses what we term relative negative correlation and positive correlation.

strong RES-E requirements reduces the electricity price by 10%, while the price would increase by 50% if a strong limit was enforced on emissions instead.⁷

Unger and Ahlgren (2005) analyse the effects of introducing a common Nordic TGC scheme and an ETS. They use the energy-system model MARKAL to simulate the electricity supply systems in the Nordic countries and show that the introduction of TGC quotas above 10% reduces the wholesale electricity price. In contrast, retail electricity prices could increase or decrease, depending on the level of the RES-E quota, than if RES-E (TGC) quotas were absent.⁸

- (iii) Some studies have specifically focused on the interactions between RES-E support and electricity prices. However, the empirical literature is inconclusive on the net effect, showing either an increase or a reduction in final consumer costs.

Concerning the former, Sorrell (2003) notes that RES-E policy would increase UK electricity prices by 4.4% in 2010. In turn, the German power-intensive industry has complained that power prices increased due the feed-in tariff and that this increase damaged its competitiveness, managing to get a special renewable energy mark-up fixed in absolute terms at 0.05 €/cent/kWh (Bode, 2006).

The European Commission itself has reported that, based on the assumption of a reference spot price of €48.6/MWh for electricity, final electricity prices could be 5% higher due to the additional RES-E investment in 2020 (European Commission, 2007).⁹ Finally, Eurelectric (2004) shows that existing RES-E support schemes in the EU would increase average European electricity prices in 2010 between 2.7 and 11.9 €/cent/kWh.

In contrast, other papers show that a reduction in the wholesale price may be negatively correlated with RES-E support and that this reduction may totally offset the cost of RES-E support (i.e., absolute negative correlation). For example, Rathmann (2007) shows that the additional amount of RES-E supported by the German RES-E policy (EEG) reduced the wholesale price of electricity in 2005–2007 by 6.4 €/MWh, while increasing the RES-E fee by 3.8 €/MWh. Thus, without the RES-E support, the retail price of electricity would have been 2.6 €/MWh than it actually has been.

Bode (2006) argues that the German FIT probably led to a reduction of wholesale electricity prices and final electricity costs.¹⁰ If and to what extent power costs decrease due to the support scheme depends on the specific characteristics of the market and the value attached to the “greenness” of power from RES-E.

In turn, Martin (2004) estimates the impact of PV generation on prices with a model of the regional US power markets, focusing on the case of New England. It is found that, while the average wholesale price in New England for 2002 was 3.78 ¢/kWh, 1 GW of PV would have reduced the average

wholesale price to 3.71 ¢/kWh, a 2% saving. This would amount to savings of \$129 million annually.¹¹

With the help of the models developed in the EU projects Green-X and FORRES 2020, Ragwitz et al. (2006) observe that transfer costs for consumers (due to the promotion of RES-E) and electricity prices are negatively correlated. They find evidence of a *relative negative correlation*, i.e., the consumer burden due to higher costs of RES-E promotion would be (only partly) compensated by lower wholesale electricity prices.

3. Analytical framework

It is assumed that wind electricity production puts a downward pressure on the wholesale market price of electricity as a result of two effects:

- First, by displacing the marginal (thermal-based) conventional technology, which eventually, would have set the market price, wind electricity has a dampening effect on this price (*direct effect*). Since the conventional technology has a higher marginal cost than wind generation, which has very low variable costs, this displacement leads to lower electricity prices compared with the absence of wind generation. This effect would disappear in the medium/long term once a new equilibrium was reached.¹²
- On the other hand, wind electricity generation leads to a lower demand for conventional electricity and, thus, to a reduction in the demand (and a lower price) for CO₂ allowances in order to comply with the EU ETS targets. This *indirect effect* would result in a lower electricity price.¹³ In contrast to the direct effect, however, this indirect effect would continue to be felt in the long term, since it affects the relationships between the costs of the different conventional technologies.

The rest of this section further analyses these two effects from a theoretical perspective.

3.1. The direct effect

In order to analyse the direct effect, the price-setting mechanism in the Spanish electricity market should be briefly described.¹⁴

A key distinction in the Spanish electricity system is that between “total electricity demand” and “residual thermal demand”. The former refers to the amount of electricity demand in each hour during the year. The so-called “residual thermal demand” is the result of subtracting the “non-manageable” generation (mostly RES-E such as wind, hydro and solar) from the total electricity demand in each hour, i.e., the “residual thermal demand” is the electricity demand, which is covered with thermal generation and it is the one responsible for the setting of electricity prices in the wholesale market.

⁷ The results of the model confirm some of the relationships analysed in Jensen and Skytte (2003), particularly with respect to the reduction in the spot market price of electricity as a result of strong RES-E requirements.

⁸ With high TGC quotas (i.e., above 30%), retail electricity prices would be higher than if TGC obligations were absent. In this case, the relatively high TGC price would overshadow the reduction in the wholesale electricity price.

⁹ This is so even if this document admits that a downward pressure on wholesale electricity prices is possible: “Marginal costs of renewable energies are often low compared with conventional energy sources, and therefore a gradual increase in renewable energies in the wholesale electricity market will reduce the wholesale market prices of electricity. The net effect on power costs to consumers is thus constituted of two counteracting effects” (European Commission, 2007, p. 17).

¹⁰ The author provides a simple numerical analysis assuming “a synthetically but close to reality power market”.

¹¹ The authors show that adding more PV generation saves price spikes and the savings increase accordingly, but even a small amount of PV can have significant impact: in the “conservative estimate” scenario 50 MW PV (0.15% of total capacity) would save \$25 million or 0.5% in the power market, increasing to savings of \$199 million or 4.1% for a 2 GW PV capacity (6% of total capacity).

¹² In equilibrium, the market price only depends on the relationship between the costs of the different conventional (thermal-based) technologies, which is unaffected by the penetration of wind electricity (see below).

¹³ It is assumed that, although CO₂ allowances are allocated freely in the EU ETS, using them to comply with emissions has an opportunity cost, which is internalised in the cost of electricity production.

¹⁴ See Crampes and Fabra (2005) and Pérez Arriaga (2005) for further details.

All thermal plants bid their generation in a given hour taking into account their own (opportunity) costs. A market price is thus set for every hour of the year in the day-ahead market and is equal to the marginal cost of the marginal thermal technology, as revealed in the bidding procedure with the participation of all the thermal technologies.¹⁵ The annual average market price of electricity is equal to the average market price of electricity in every hour within a year (i.e., 8760 h) weighted by the level of electricity demand in those hours.

The marginal technology is determined by the level of electricity demand and the costs of thermal technologies. Thus, the electricity supply curve is built in each hour accordingly:

- When demand is very high, the price in this hour is set by the costs of a high marginal cost thermal generation technology, i.e., fuel would be the marginal technology.
- When demand is very low, the price in this hour is set by the costs of a cheap thermal technology (i.e., nuclear).

All this is shown in the upper part of Fig. 2, which represents the coverage of residual thermal demand in each of the 8760 h of a year (i.e., the monotonous curve of thermal demand), ordered from left to right according to the level of residual thermal demand, i.e., from higher (near the intersection of the x - and y -axis) to lower demand levels. It is clear that the hours of greater thermal demand (i.e., those on the left) need to be covered (in the margin) with the highest marginal cost technologies (i.e., fuel), whereas the hours of lower demand can be met with low marginal cost plants.

The analysis of the direct effect is based on the assumption of a perfectly competitive electricity system, which is in equilibrium and, according to the conditions in the Spanish electricity market, exclusively based on thermal technologies (i.e., no wind generation).¹⁶

More specifically, we assume that there are four thermal-based generation technologies available: nuclear (N), coal (C), gas (G) and fuel (F). Their total costs per MW of installed capacity from 0 to 8760 h are represented in the intermediate part of Fig. 2. Each of them has a fixed cost (given by the intersection of the curves and the y -axis) and a variable cost (given by the slope of the curve). For simplicity and clarity of exposition, we have assumed that the cost curves are linear. This implies constant marginal costs for each type of power plants, i.e., power plants within each category have the same marginal costs. The implications of this and other assumptions for the results of the analysis are discussed at the end of this section.

The relationship between the costs of the thermal technologies determines the hours of operation of those technologies as well as the time of the day when they set the electricity price. The intersection of their cost curves determines the number of hours

in which the operation of each technology is optimal (i.e., the one that minimises the total costs for the system). According to Fig. 2:

- F is the optimal technology up to f hours of operation.
- G is the optimal technology for g to f hours of operation.
- C is the optimal technology for c to g hours of operation.
- N is the optimal technology for a number of hours of operation above c .

Therefore, by taking into account how electricity demand is met hour-by-hour, it is possible to identify the optimal capacity installed in the system for each type of thermal technology when the system is in equilibrium (upper part of Fig. 2).¹⁷ The lower part of Fig. 2 shows that the market price is set by the marginal cost of the marginal technology, where MC_i represents the marginal cost of technology i .

What would happen to this system in equilibrium, if previously absent wind generation entered the system? This new wind generation would replace thermal-based electricity production, leading to the following results (Fig. 3)¹⁸:

- Given its stochastic nature, the expected wind generation in a given hour (MWh_E) depends on its specific probability distribution, between a minimum (cero) and a maximum value (corresponding to the installed wind capacity, MW_E). Both extremes of the continuum would have a low probability of occurrence.
- Given the new wind production, the residual thermal demand would change, fluctuating between those two extremes. We would then have:
 - A monotonous curve of maximum residual thermal demand (with minimum wind generation, i.e., $MWh_E = 0$ in all hours).
 - A monotonous curve of minimum residual thermal demand (with maximum wind generation, i.e., $MWh_E = MW_E$ in all hours).
 - A monotonous curve of expected residual thermal demand (if wind generation equals $E(MWh_E)$ in all hours).¹⁹

The reduction of the residual thermal demand due to wind generation leads to a shift in the electricity demand curve down and to the left (upper part of Fig. 4). In turn, this reduces the need for high marginal cost thermal generation and demand is thus more likely to be covered with generation technologies with lower marginal costs, i.e., the most expensive technologies operate fewer hours and the cheapest (base-load) technologies operate longer. Therefore, the technologies with a higher marginal cost

¹⁵ "The day-ahead market is made up of 24 hourly markets that clear once a day. Electricity producers submit supply functions specifying the minimum price at which they are willing to produce a given amount of output from each of their plants. Actors on the demand side submit demand functions specifying the maximum price at which they are willing to purchase a given amount of electricity. The market operator thus constructs a merit order dispatch and the equilibrium prices are determined through market clearing (i.e., intersection between the supply and demand curves). Conditionally on being dispatched, the price to be received by producers is set according to a uniform-price auction. Irrespective of their bids, the price received by producers is set equal to the highest accepted supply bid (the so-called Marginal Price) (Crampes and Fabra, 2005, p. 130). This highest bid corresponds to the marginal technology".

¹⁶ Before wind generation enters the system, it is assumed without loss of generality that the mix of generation capacity is in equilibrium and is optimal. This means that this is the mix, which leads to a lower electricity price subject to the constraints that the investment costs of all technologies are recovered (paid-off).

¹⁷ For simplicity and clarity of exposition, non-supplied energy (NSE) is not represented in the figure. NSE would be similar to a thermal technology with a zero fixed cost and a variable cost equal to the valuation of energy by the consumers. This valuation is normally above the highest variable cost of the existing thermal technologies. Thus, in an system in equilibrium, it could be expected that there was a (very small) number of NSE hours, determined by the intersection between the NSE curve and the curve of the optimum thermal technology for the lowest possible number of hours (i.e., technology F in the figure).

¹⁸ The analysis would be similar if an increase in wind generation was considered (i.e., if there was some amount of wind generation initially).

¹⁹ For simplicity of exposition, the figure assumes that the probability distribution of wind generation is the same in all the hours of a year. Therefore, it has the same expected value at any time during the year. In Spain, wind generation tends to be slightly higher in the hours with a lower price, i.e., wind generation would tend to be concentrated in the hours with a lower demand. This would not have any impact on the new equilibrium situation created by wind generation, although it would have an impact on the transition until a new equilibrium is reached.

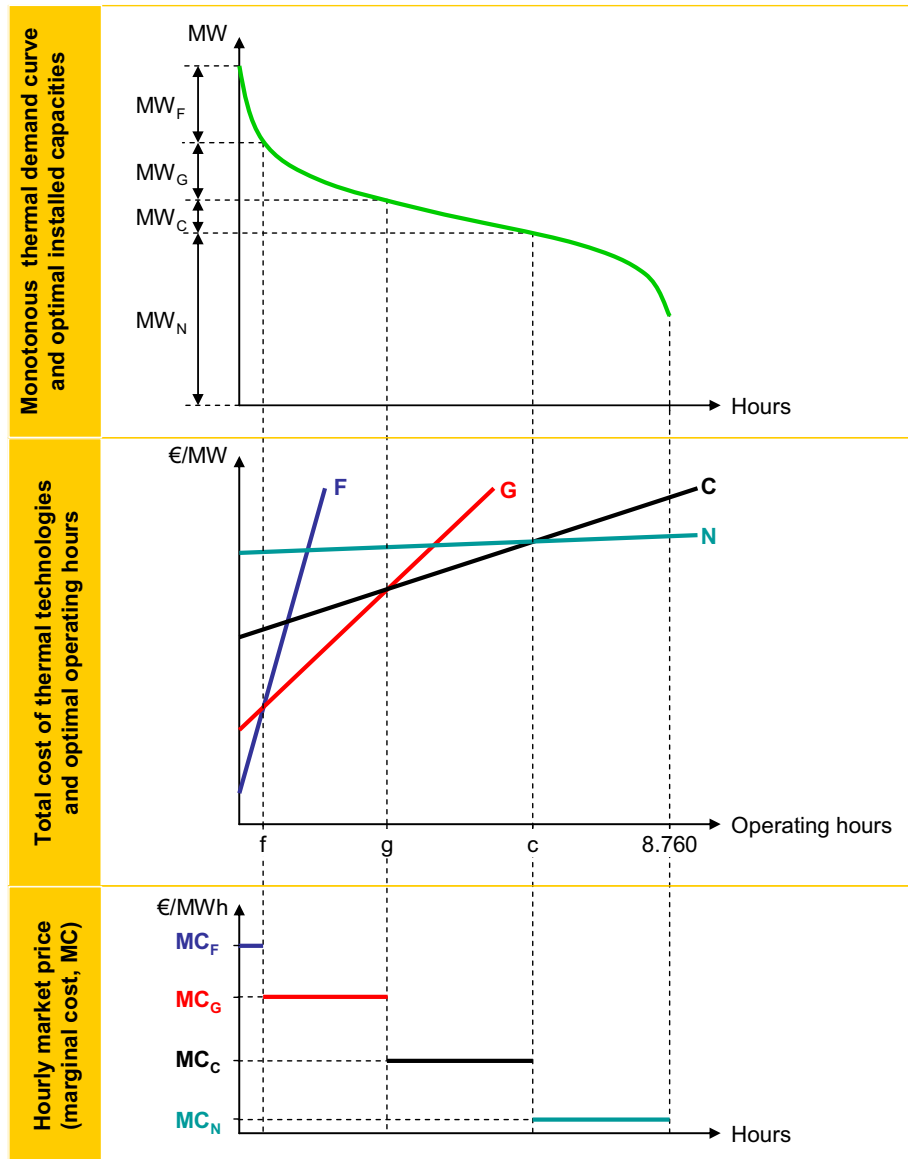


Fig. 2. Initial equilibrium: installed capacity of thermal technologies and hourly market prices.

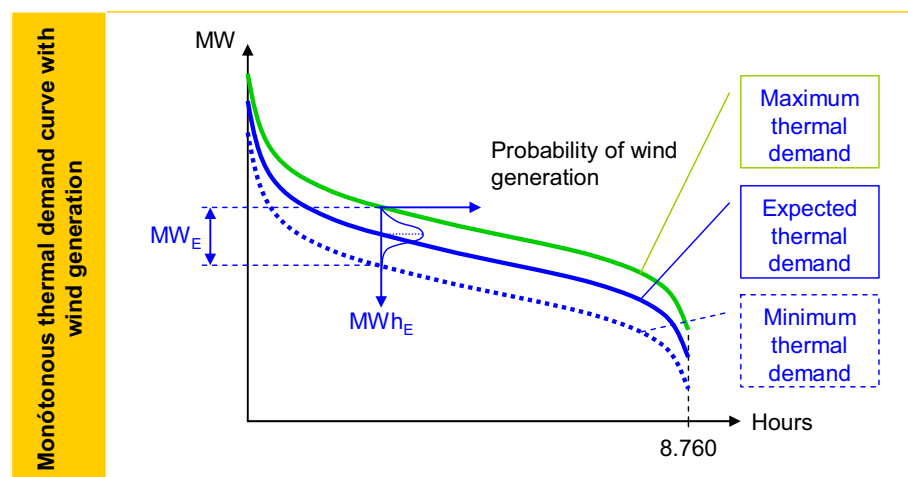


Fig. 3. Introducing wind electricity in the electricity market.

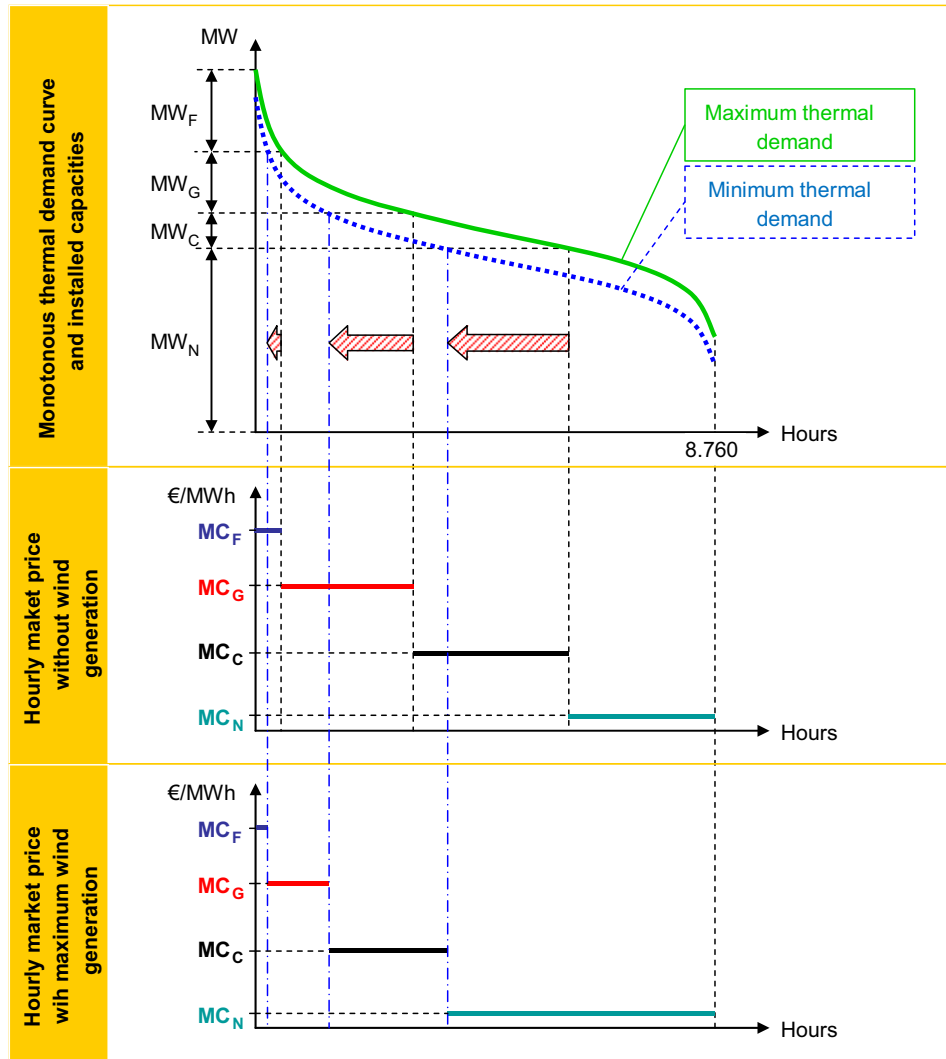


Fig. 4. The impact of introducing wind electricity on electricity prices.

(i.e., fuel) will set the price in a fewer number of hours compared with a situation without wind generation.

By comparing the intermediate part of Fig. 4, which represents the marginal costs of the thermal-based technologies operating before the penetration of wind electricity, with the lower part of this figure, which shows the marginal costs after the introduction of wind generation, it can be seen that the marginal cost segment of the lowest-marginal cost technology (nuclear) is longer, whereas the segments for the expensive thermal-based generation technologies (particularly fuel and gas) are shorter. This immediately reduces electricity prices, which would then be lower than without wind generation.

The reduction of residual thermal demand and the new wind capacity leads to a surplus of base-load capacity in the system, although base-load plants will not close down, because the electricity price is above their very low short-term marginal costs.

However, this new situation with lower electricity prices, a shortage of peak capacity with respect to the existing capacity before wind generation ($MW'_F > MW_F$) and a surplus of base-load capacity ($MW'_N < MW_N$) is only temporary and will not last much. In so far as base-load plants reach the end of their useful lives and electricity demand increases in the future, there will be a need for new capacity. These factors will lead to the construction of new

peak-load plants, a reduction in the shortage of this type of plants, a reduction in the surplus of base-load plants and a gradual increase in electricity prices, until the pre-existing equilibrium shown in Fig. 2 is recovered (Fig. 5). It can be observed that, in this new equilibrium, the market price (given by the marginal costs of the technologies in the lower part of Fig. 5) is likely to be similar to the one which would result from the absence of wind generation (see the lower part of Fig. 2 or the intermediate part of Fig. 4), as the imbalances created by the penetration of wind generation are corrected and the final equilibrium is reached.

Furthermore, the new situation shown in the upper and lower parts of Fig. 4 is not a long-term equilibrium because, in equilibrium, the operating hours of each thermal technology (and, thus, market prices) only depend on the cost comparison between the thermal technologies and this comparison is not affected by a greater wind generation, i.e., the points of intersection between the cost curves of the thermal technologies are unchanged. This is why the intermediate parts of Figs. 2 and 5 are very similar (i.e., cost curves intersect at similar points).

However, reaching the new equilibrium will take time, mostly depending on the rate of demand growth and, thus, electricity prices will be lower in the meantime. Thus, the speediness with which the system reaches a new equilibrium will determine the

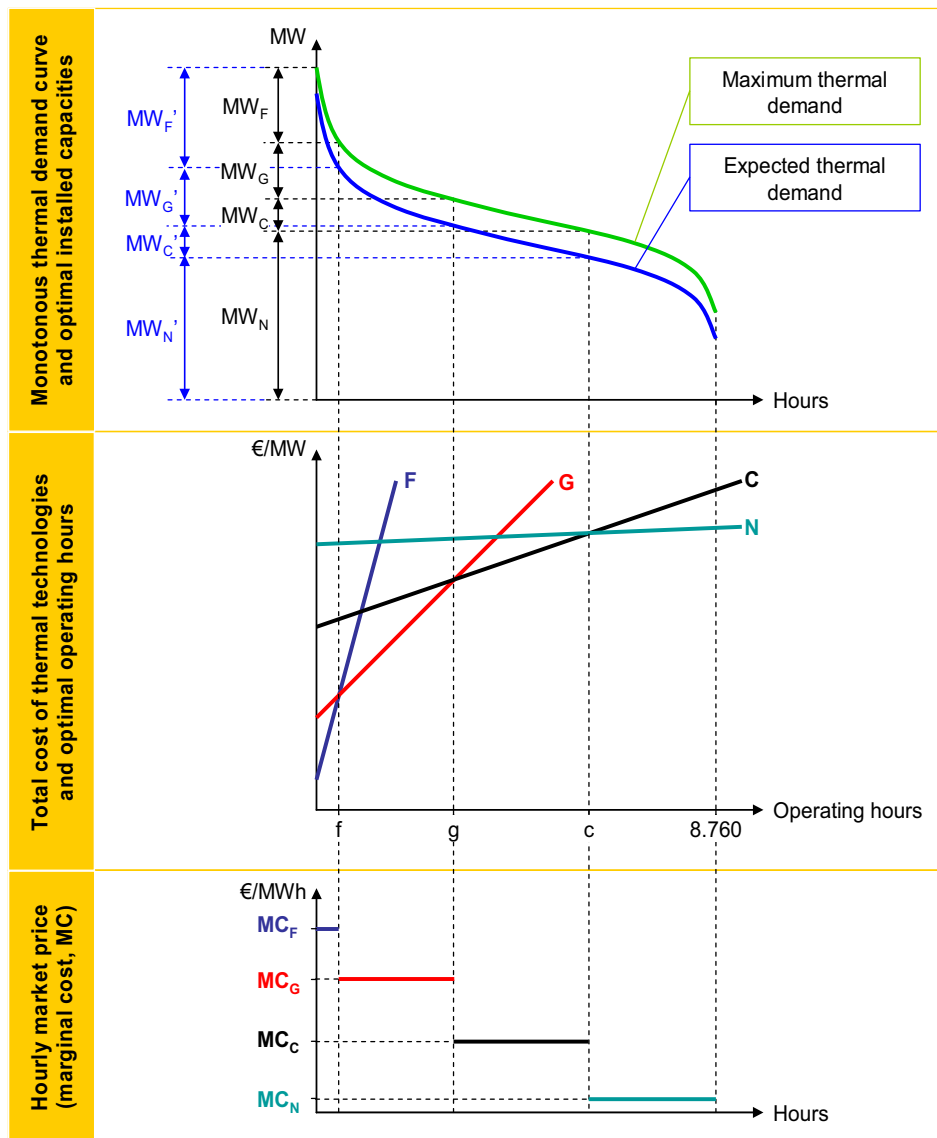


Fig. 5. The new equilibrium in the electricity market as a result of wind electricity generation.

extent to which wind generation is able to reduce the market price, i.e., the direct effect is only temporary. This speediness will depend on several factors, including:

- *Demand growth rates.* The faster the growth rate of electricity demand, the easier this increased demand will absorb the excess base-load capacity resulting from the new equilibrium.
- *The remaining useful life of the existing base-load capacity.* If the surplus thermal base-load capacity (in the new equilibrium) is paid-off in the short-term and this capacity is retired, the new equilibrium will be reached faster.
- *The magnitude and the speed of penetration of wind capacity.* The faster the penetration of wind capacity into the system and the greater this capacity, the greater the distance between the new and the old equilibrium and the longer it will take to reach the new equilibrium.
- *The restrictions to the deployment of new thermal capacity.* The new equilibrium will be reached in a longer time frame as a result of several constraints, including administrative procedures, access to fuels, grid connection procedures, environmental restrictions and availability of sites.

Summing up, we could distinguish the impact of wind generation on the electricity price in a pre-equilibrium (related to the short/medium terms) and a post-equilibrium (medium/longer terms) situations. Whereas the former results in a reduction in annual average electricity prices, the latter leads to similar market prices to those observed in the absence of wind generation.

It should also be noted that the necessary total thermal capacity does not change when wind generation is introduced, because of security of supply concerns. Since wind production could well be zero, the thermal installed capacity with or without wind capacity should be the same. However, a greater wind capacity in the system does not increase the need to invest in additional thermal capacity in order to maintain the same security of supply, as argued by some commentators. As long as a given installed wind capacity has a high probability to produce a minimum quantity of electricity, additional investments in thermal capacity are not needed, leading to a saving for the system.

3.2. Indirect effect

Thermal-based plants in the EU are covered by the EU ETS, i.e., they have to handle CO₂ allowances for their emissions. Thus, a

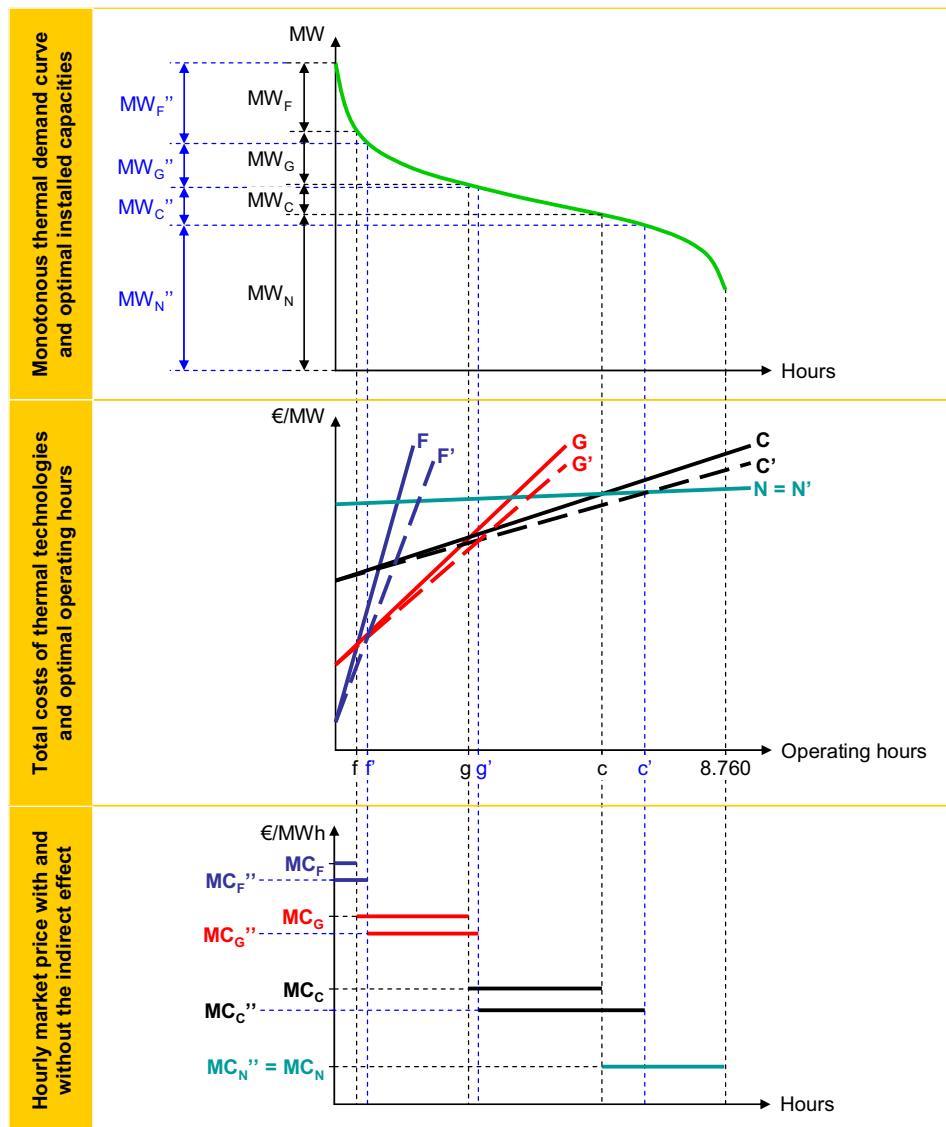


Fig. 6. The impact of the “indirect effect” on the electricity market.

second impact of wind generation on the electricity price is related to the reduction in the demand for CO₂ allowances, which are necessary to comply with the EU ETS targets and a lower price for those allowances (given the lower demand for them). Thus, the substitution of wind for conventional electricity would lead to a double cost saving for electric utilities.²⁰ The aforementioned price effect is the result of:

- The reduction in thermal production, which leads to a lower demand for CO₂ allowances and, thus, to a lower CO₂ allowance price.²¹
- Since CO₂ allowances represent an opportunity cost for conventional electricity generators, which is internalised in

their electricity bids, a reduction in the demand and price of CO₂ allowances would lead to a reduction in the electricity price.

This effect would affect the electricity market equilibrium, because changes in the CO₂ allowances price also change the relationships between the costs of the different thermal technologies (the new costs of the thermal technologies with this indirect effect are represented by the F' , G' , C' and N' curves, see Fig. 6), and, thus, the operating hours of each thermal technology in the equilibrium (f' , g' and c'). In contrast to the direct effect, the intersection points between different technologies are defined.

In turn, this change in the number of operating hours of each thermal technology leads to a change in the thermal generation capacity in equilibrium, as shown in the upper part of Fig. 6. It can be observed that, in this new equilibrium, the capacity of base-load plants would decrease ($MW_N'' < MW_N$) and peak capacity would increase ($MW_F'' > MW_F$). In other words, the reduction in the CO₂ allowance price as a result of wind generation leads to an increase in the number of hours of operation of the most expensive thermal plants.

²⁰ Of course, there could also be a double revenue loss for utilities if RES-E plants are set up by other players. We thank an anonymous referee for this remark.

²¹ In case additional RES-E generation only served to meet additional electricity demand, one would not expect any change in the CO₂ market, i.e., CO₂ prices (and power prices) would remain constant. Therefore the price reduction in Fig. 6 occurs with respect to either a case where additional demand would have been met by conventional sources or if the increase in RES-E generation is lower than the increase in electricity demand. We thank an anonymous referee for this remark.

Regarding the impact of this effect on the electricity price in the new equilibrium (lower part of Fig. 6), two opposing effects can be distinguished:

- On the one hand, the short-term reduction in the variable costs of thermal generators (for example, $MC'_F < MC_F$) as a result of a lower demand for allowances would lead to a lower average market price.
- On the other hand, the increase in the number of operating hours of the most expensive thermal technologies would tend to increase the average market price. The latter medium/long-term effect would not offset the aforementioned short-term reduction. Therefore, the net impact of both effects would be a reduction in the market price.

Table 2 discusses the implications for the results of the most relevant assumptions adopted in this theoretical part.

In order to analyse the impact of a greater RES-E generation on the final consumer, the positive impact on the wholesale electricity price (direct and indirect effects) should be compared with the costs of RES-E support because either an increase or a reduction in (retail) electricity prices may result. This empirical issue is explored in Section 5 for wind generation in Spain.

Table 2
Impact of the assumptions on the results of the analysis

Assumptions	Major implications
Linear cost curves (i.e., constant marginal costs for each type of power plants)	In the real world, the marginal costs of plants of the same type are different. This is a common simplifying assumption used in the studies on interactions (see Bode 2006 for a similar assumption) and does not affect the conclusion that wind generation would displace thermal generation out of the market, although, of course it would affect the conclusion that all plants within a (high marginal cost) thermal generation category would be driven out of the market. In reality, some plants within a category would be displaced (those with the highest marginal costs) while others (the cheapest ones) would not. The conclusion that there could either be an absolute or a relative negative correlation due to RES-E generation would not be affected by this assumption.
Perfect competition and marginal cost pricing. The market price is set by the marginal cost of the marginal technology	In a case of imperfect competition, marginal cost pricing would be unlikely, i.e., prices would tend to be above the marginal costs of the marginal technology. The displacement of thermal-based technologies as a result of a greater RES-E generation would be less likely. Thermal plants with a marginal cost above the perfect competition electricity price and below an eventual imperfect competition electricity price would not be driven out of the market in this case, i.e., a lower RES-E generation than in the case of perfect competition would result and, thus, the impact on the electricity price would also be smaller. Or it would be necessary to provide a greater support level for RES-E plants in order to increase their penetration in this imperfect electricity market. Therefore, with imperfect competition, absolute and relative negative correlation would be less likely, and positive correlation would be more likely

Table 2 (continued)

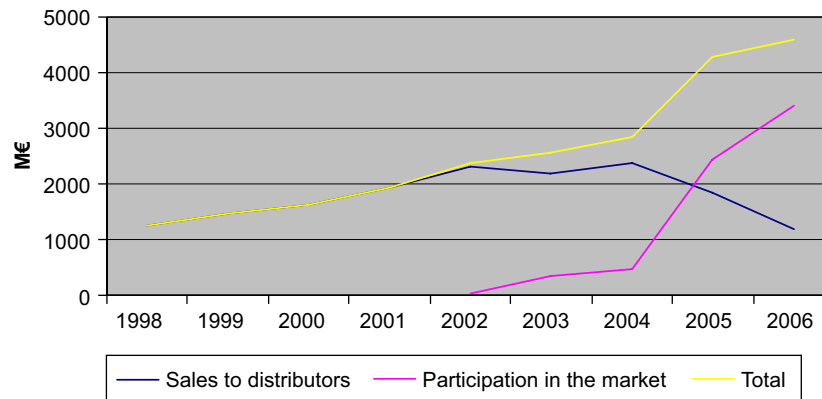
Assumptions	Major implications
Lower marginal costs of RES-E (wind) generation	than with perfect competition: the positive direct effect of RES-E generation on power prices would be less likely to occur with imperfect competition or it would be smaller than with perfect competition. RES-E generation is traditionally considered to have low marginal costs, mostly due to its relatively low O&M and fuel costs. However, if those costs were higher than generally considered, RES-E generation would have more difficulties to displace conventional generation (or a greater amount of support would be necessary in order to encourage it). In turn, this could lead to a lower direct effect on the electricity price or the higher support would make absolute and relative negative correlation less likely than if low marginal costs for RES-E were assumed.
Opportunity costs of freely allocated CO ₂ allowances	It is assumed that, although CO ₂ allowances are allocated freely in the EU ETS, using them to comply with emissions has an opportunity cost which is internalised in the cost of electricity production. The substitution of RES-E for thermal-based electricity reduces the amount of CO ₂ allowances needed by the electricity sector to comply with its emissions targets, which tends to put downward pressure on its price. However, if freely allocated CO ₂ allowances are not internalised as an opportunity costs, the indirect effect would be less likely to happen. This is so because the reduction in their price (as a result of a greater RES-E generation) would not be considered a cost saving by the electricity companies and, thus, the substitution of RES-E for fossil-fuel electricity would not affect electricity prices.
Non-supplied energy (NSE) is not considered.	NSE would be similar to a thermal technology with a zero fixed cost and a variable cost equal to the valuation of energy by the consumer. This valuation is normally above the highest variable cost of the existing thermal technologies. Thus, in an system in equilibrium, it could be expected that there was a (very small) number of NSE hours, determined by the intersection between the NSE curve and the curve of the optimum thermal technology for the lowest possible number of hours (i.e., technology F in Fig. 2). NSE is normally very small and, thus, its non-inclusion has a negligible impact on the conclusions.
The probability distribution of wind generation is the same in all hours of a day	This means that wind generation would have the same expected value at any time during the year. The experience in Spain shows that wind generation tends to be slightly higher when the electricity price is low (i.e., at times of lower electricity demand). Wind generation would tend to drive conventional generation out of the market at times of low electricity demand and high wind generation. Therefore, the direct effect of wind generation would be less likely at times of high electricity demand. This would not affect the new equilibrium situation created by wind generation, although it would have an impact on the transition until a new equilibrium is reached.

Table 3

Support (tariffs and premiums) for wind on-shore electricity according to RD 661 (cents/kWh)

	Period	Regulated tariff	Reference premium	Upper limit	Lower limit
Wind on-shore	First 20 years	7.3228	2.9291	8.4944	7.1275
	After 20 years	6.1200	0.0000	–	–

Source: Own elaboration from RD 661.



Source: CNE (2007a).

Fig. 7. Evolution of the total costs of RES-E support in Spain. Source: CNE (2007a).

4. Wind electricity and its promotion in Spain

This section provides a brief overview of the trends in wind electricity generation in Spain and describes the main features of the public promotion scheme applied in this country.

4.1. Wind electricity generation in Spain

With 11,728 GW in 2006, Spain was the second country in the world in terms of wind-energy-installed capacity, only behind Germany (18 GW) (2005 data). The growth rates of wind-installed capacity in the last decade in this country are impressive. In the 1998–2005 period, wind capacity has grown at an average annual rate of 42%.

Wind electricity generation in 2006 reached 22,941 GWh and currently covers 10% of electricity demand in Spain (CNE, 2007a).

A great wind electricity potential, a favourable regulation, a high maturity and technological evolution and a great manufacturing capacity of the domestic industry are factors behind this success (IDAE, 2005). More than 70% of wind capacity installed by the end of 2005 was supplied by Spanish manufacturers.

4.2. Public promotion of wind electricity in Spain

Following European guidelines and legislation (including the 1997 White Paper on Renewable Energy Sources²² and Directive 77/2001/EC on renewable electricity), but also as a result of an internal acknowledgment of the socioeconomic and environmental benefits of RES for the country, renewable energy promotion has been a national policy priority in Spain for more than a decade. This has come specifically to a detailed renewable energy plan (in 1999, updated in 2005, see MITC, 2005), a feed-in tariff system for RES-E and a commitment by all political parties to

maintain the system and avoid discontinuities in support, which negatively affects investment in renewables.

Concerning RES-E promotion at the national level, the feed-in tariff (FIT) scheme was implemented in 1998 (Royal Decree 2818/1998), and has been subsequently modified in 2004 (Royal Decree 436/2004) and, recently, in June 2007 (Royal Decree 661/2007).

The RES-E generator has two options:

- He can sell his electricity to the distributor at a regulated tariff.
- He can sell his electricity directly in the power market. He would receive the price negotiated in the market plus a premium.

In case (b), maximum and minimum profitability thresholds are introduced for some technologies, including wind electricity. This “cap-and-floor” system introduced by the Royal Decree 661/2007 is a particular design feature of the system with respect to other FITs in the world. The upper threshold limits windfall profits for RES-E generators and ensures that consumers will not be overburdened with the costs of RES-E promotion. The minimum threshold encourages investments in RES-E by reducing the risks for investors. Table 3 shows the support levels for wind electricity under both options (a and b).

Fig. 7 shows the evolution of the total costs of RES-E support between 1998 and 2006, of which wind generation represents the lion's share. This support amounted to 4582 M€ in 2006. However, this figure is biased, since it does not take into account the reduction in the price of electricity due to a greater RES-E generation. Calculating this reduction is the aim of Section 5.

5. The empirical analysis

Carrying out a quantitative analysis of the impact of the aforementioned direct and indirect effects on the electricity price

²² White Paper for a Community Strategy and Action Plan “Energy for the future: Renewable Energy Sources of Energy” COMM(97)599 final.

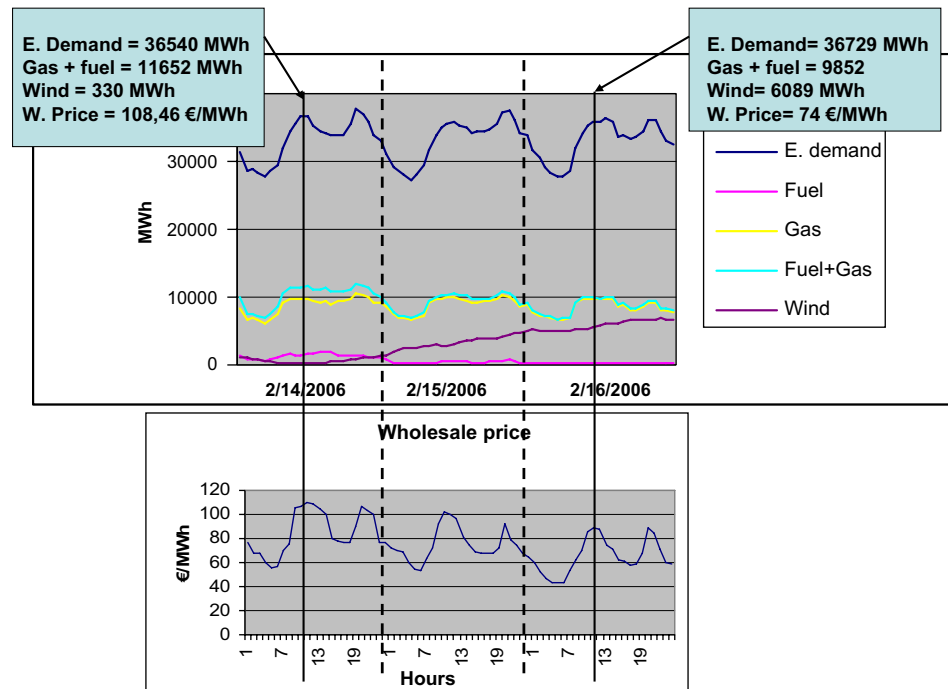


Fig. 8. Illustrating the impact of a greater wind generation on electricity prices in Spain.

in the Spanish context is a highly complex task at least for the following reasons:

- Stochastic character of wind generation, which makes it difficult to identify its appropriate distribution.
- Possible correlation between wind generation and generation from other RES (i.e., hydro), which could bias the estimate of the residual thermal demand.
- Availability of information regarding some relevant factors, with a potentially high impact on the market price (for instance, performance of thermal plants, availability of these plants, etc.).

Given these and other issues, only the direct effect is empirically assessed in this paper. The following two analysis are proposed:

- Identification of the direct effect through the use of historical data (Section 5.1).
- Identification of the direct effect by simulating what would have been the merit order (and, thus, the market price) in the absence of wind generation. This would allow us to calculate the cost savings for the system as a result of a lower electricity price due to wind generation (Section 5.2).

5.1. Identification of the direct effect: historical data

A simple approach to identify the direct effect is to analyse the existing correlation between the market price and the level of wind production. A priori, as argued in the analytical framework at least for the short/medium term, it could be expected that a greater wind generation would reduce the residual thermal demand, drive the most expensive thermal technologies out of the market and, thus, lead to a lower market price. However, modelling such relationship through a simple regression with those two variables could lead to misleading results. This is so because several factors, apart from wind generation, influence the market price, including the level of electricity demand, the level of hydro generation, maintenance activities, the unexpected unavail-

ability of thermal plants (due to breakdowns) and the evolution of the costs of the different fuels.

Therefore, we analyse the market in 3 consecutive days with similar levels of electricity demand in order to isolate the impact of wind generation from the other factors affecting the market price. Albeit simple, this methodology mitigates the distortionary impact stemming from changes in: (a) the level of demand; (b) the level of fuel and CO₂ prices; (c) planned maintenance activities in the plants; (d) differences in the levels of electricity generation with hydro sources.

The days chosen for the analysis were 14–16 February 2006. These were working days (more specifically, a Tuesday, a Wednesday and a Thursday).

In order to identify the influence of wind generation on the electricity price, Fig. 8 compares the same hours of 2 days with similar levels of electricity demand, i.e., between 10 a.m and 1 p.m. of 14 and 16 February:

- On 14 February there was almost no wind generation in the aforementioned hours, while fuel and gas generation was significant. The operation of fuel plants, the most expensive in the system, led to a market price of 110 €/MWh.
- On 16 February, a substantial amount of wind generation in the same hours can be observed. Fig. 8 shows that, with the same level of electricity demand, a greater amount of wind generation drives the most expensive thermal generation (fuel and gas) out of the market, i.e., they are no longer needed to cover electricity demand. This reduces the electricity market price from around 110 €/MWh (on the 14th) to 75 €/MWh (on the 16th). Indeed, a high negative correlation coefficient (i.e., -0.9) between the increase in wind electricity and the reduction of electricity prices between 10 a.m. and 1 p.m. in day 16th can be observed.²³

²³ However, the correlation coefficient between wind generation and market prices for the whole period (72 h in the aforementioned 3 days) was very low (-0.42). A simple econometric model estimated with ordinary least squares taking market price as the dependent variable and wind generation as the single

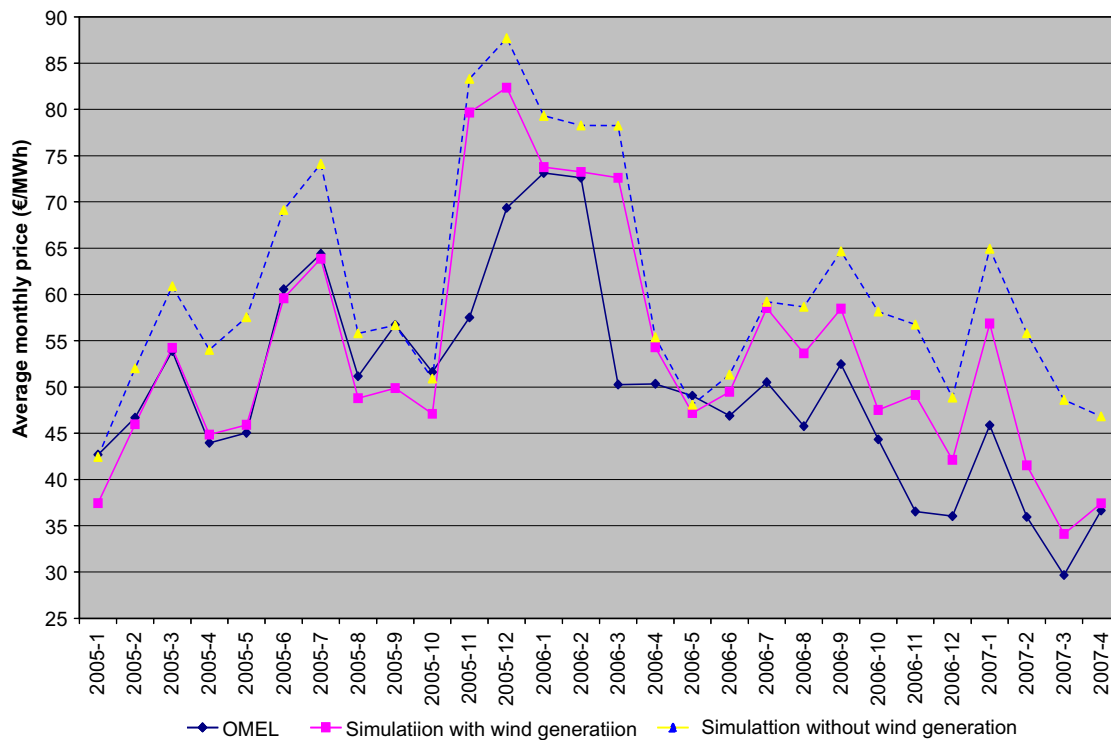


Fig. 9. Simulating the impact of a greater wind generation on electricity prices in Spain.

Since the cost of support for wind generation in 2006 was 39 €/MWh²⁴ and the resulting reduction in the electricity market price was 35 €/MWh, we can conclude that *relative negative correlation* occurred (see Section 2.1), i.e., that the reduction in the wholesale price of electricity did not offset the costs of RES-E support, increasing the retail price.

5.2. Simulating the direct effect

An alternative approach to quantify the direct effect was to simulate the merit order (dispatch) and the electricity price in 2005, 2006 and in the first 5 months of 2007 in the absence of wind generation (i.e., 29 months were analysed). The direct effect is estimated as the difference between the prices simulated with and without wind generation.

Simulating the dispatch is a highly complex exercise, given the numerous economic and physical variables and the restrictions affecting this dispatch. At least the following variables, and their possible trends, should be considered: restrictions related to ramps, number of stops/starts per type of thermal plant, which can reasonable be expected annually, availability of the plant, minimum technical production, hydro production as a function of new market prices, price-elasticity of demand and impact of changes in market prices on imports and exports. The consideration of all these variables, restrictions and relations in the

simulation, although desirable, is beyond the scope of this analysis. We have opted for a more simple simulation methodology with the following assumptions:

- Restrictions due to ramps and those associated to the number of stops/starts that may reasonably occur in a plant during a year are not considered.
- The rates of non-availability of technologies assumed in the simulation are those actually observed in the 29 months used for this analysis.
- The dispatch of hydro plants, which has been assumed is the one actually observed during this period.
- The demand for electricity in the system is totally inelastic.
- Imports and exports are those actually observed in the period.

Obviously, the change in these simplifying assumptions would affect the market price resulting from the simulation whose evolution, therefore, could not be attributed entirely to the increase in wind generation. For example, for the same level of wind generation, a greater number of non-expected stops in thermal plants would affect the merit order of thermal technologies and, thus, market prices, irrespective of the influence of wind generation.

The following sources of data have been used for this analysis:

- Data on thermal capacity installed, monthly non-availabilities of thermal plants, thermal production (coal, gas and fuel) and wind generation have been obtained from Red Eléctrica de España (REE).
- International fuel markets: The price for gas in the UK National Balancing Point and the CO₂ allowance price in the European Energy Exchange (EEX) have been used to estimate the generation costs of thermal plants.

Fig. 9 shows the evolution of the average monthly prices actually observed in OME (the Spanish electricity market operator in the period), the average monthly prices corresponding

(footnote continued)

regressor (i.e., no control variables) led to a very low adjusted R^2 (0.17) but the explanatory variable is significant at the 95% confidence level (t -statistic: -3.93), i.e., wind generation is one of the factors (although certainly not the only one, as mentioned above), which explains the evolution of electricity market prices.

²⁴ Own calculation. According to the RD436/2004 wind generation receives a support of 50% of the average electricity tariff (40% premium+10% incentive to participate in the market). The RD 1556/2005 set the average electricity tariff for 2006 at 7.6588 €/cents/KWh. Therefore, support amounts to 39 €/MW in that year.

to the simulated dispatch taking into account the real wind generation (“simulation with wind generation”) and the average monthly prices corresponding to the simulated dispatch in the absence of wind generation (“simulation without wind generation”).

The market prices in the simulated dispatch with wind generation reasonably fit those actually observed in OMEL. Although the aim of this analysis is not to replicate the real dispatch, the fact that the prices resulting from the simulation are similar to those actually observed validates the method used to estimate the impact of wind generation on the electricity price.

According to this methodology, and as shown in Table 4, wind generation would have led to a reduction in the wholesale electricity price of 7.08 €/MWh in 2005, 4.75 €/MWh in 2006 and 12.44 €/MWh between 1 January and 31 May 2007 (i.e., a reduction of 11.7%, 8.6% and 25.1%, respectively). Therefore, we conclude that the change in the merit order caused by wind generation has reduced the electricity price.

Taking into account the reduction in the electricity prices as a result of wind generation, we can thus calculate the total cost savings for consumers, which are the result of multiplying the total electricity demand in a given year by the reduction in the electricity price due to the direct effect (last column of Table 5). Substantial cost savings for the consumers can be observed.

However, in order to calculate the net financial burden for consumers resulting from wind generation, the cost savings in Table 5 should be compared with the costs of support for wind generation, which are finally paid by consumers. This is shown in Table 6.

The savings in Table 5 are subtracted from the total amount of public support for wind generation in order to calculate the net costs of wind electricity promotion in 2005–2007. Once the electricity price reduction is considered, wind electricity promotion does not entail an additional cost for consumers in the short/medium term (Table 6, last column). On the contrary, there has been a significant saving of more than 2000 M€ over the period (i.e., absolute negative correlation).

6. Concluding remarks and policy implications

The literature on the interaction between ETS, RES-E promotion and electricity prices has been rather theoretical. It has shown that the simultaneous achievement of the three goals of RES-E deployment, CO₂ emissions reductions and moderate electricity prices for consumers mostly depends on the existence of a negative relationship (correlation) between an increasing RES-E generation and the wholesale electricity price. However, despite the profound policy implications of this topic, the empirical literature is surprisingly thin.

This paper has tried to fill this gap in the literature by analysing the Spanish case. It has been shown that an absolute negative correlation exists between wind electricity promotion and the wholesale market price, i.e., that the increase in the costs of RES-E support may be offset by the short/medium-term reduction in the wholesale price, leading to a reduction of retail electricity prices. The policy implications of this finding are highly relevant because it would imply that the three objectives of RES-E deployment, CO₂

Table 4
Calculating the impact of a greater wind generation on electricity prices in Spain

Year	Average price OMEL (€/MWh) ^a	Simulation with wind (€/MWh)	Simulation without wind (€/MWh)	Direct effect of wind generation (€/MWh) ^b	Relative direct effect of wind generation (%) ^c
2005	53.63	54.97	62.05	7.08	11.7
2006	50.67	56.66	61.41	4.75	8.6
2007 (January–May)	37.04	43.28	55.72	12.44	25.1

Source: Own elaboration.

^a Average market price in OMEL day-ahead market.

^b Estimated as the difference between the simulations without and with wind generation.

^c Estimated as: (Direct effect/(Price OMEL+Direct effect)).

Table 5
Calculating the total cost savings of a greater wind generation in Spain (direct effect)

Year	Total demand ^a (GWh)	Direct effect (€/MWh)	Savings due to the direct effect (M€)
2005	246,768	7.08	1746
2006	252,819	4.75	1200
2007 (January–May)	108,382	12.44	1348

Source: Own elaboration.

^a CNE (2007b).

Table 6
Calculating the final costs for consumers of a greater wind generation

Year	Wind generation (GWh) (1) ^a	Premium+incentive (€/MWh) (2) ^b	Total support for wind generation (M€) (3) = (1) × (2)	Cost savings due to the direct effect (M€) (4)	Net cost savings (M€) (5) = (4)–(3)
2005	21,173	38	805	1746	942
2006	22,940	39	895	1200	306
2007 ^c	11,539	39	450	1348	898

Source: Own elaboration.

^a CNE (2007a).

^b RD436/2004.

^c January–May.

emissions reductions and moderate consumer prices can be achieved simultaneously through an ambitious RES-E promotion policy and that RES-E support is not as burdensome for the final electricity consumer as it is usually assumed. This is a particularly relevant issue at this moment, when renewable energy targets for 2020 in the EU are under discussion, being the costs of renewables a key element of this debate. However, these results should be taken with care, given the restrictive assumptions of this exercise. Furthermore, the results reported should be considered as an illustration of the direction of the effects and as orders of magnitude, rather than as exact measures of the impact of wind generation on electricity prices.

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References

- Bode, S., 2006. On the Impact of Renewable Energy Support Schemes on Power Prices. Institute of International Economics (HWWI), Hamburg.
- Comisión Nacional de la Energía (CNE), 2007a. Informe Mensual de Ventas del Régimen Especial en España. July 2007. Available at <<http://www.cne.es>>.
- CNE, 2007b. Boletín mensual de indicadores eléctricos y económicos (para demanda total). Available at <<http://www.cne.es>>.
- Crampes, C., Fabra, N., 2005. The Spanish electricity industry: *Plus ça change....* Energy Journal, special edition, 127–154.
- Eurelectric, 2004. A Quantitative Assessment of Direct Support Schemes for Renewables, Brussels.
- European Commission, 2006. Green Paper. A European Strategy for Sustainable, Competitive and Secure Energy. COM(2006)105 final, Brussels.
- European Commission, 2007. Renewable Energy Road Map. Renewable energies in the 21st century: building a more sustainable future. Communication from the Commission to the Council and the European Parliament. COM(2006) 848 final, Brussels, 10.1.2007.
- Hindsberger, M., Nybroe, M.H., Ravn, H.F., Schmidt, R., 2003. Co-existence of electricity, TEP and TGC markets in the Baltic Sea Region. Energy Policy 31 (1), 85–96.
- IDAE, 2005. Renewable energy and energy efficiency. Bulletin no. 7. Madrid (in Spanish) <<http://www.idae.es>>.
- Jensen, S., Skytte, K., 2003. Simultaneous attainment of energy goals by means of green certificates and emission permits. Energy Policy 31, 63–71.
- Martin, K., 2004. Thought Experiment—How PV Reduces Wholesale Power Prices in New England. Photon International December 2004.
- Ministry of Industry, Tourism and Trade (MITC), 2005. Renewable Energy Plan. IDAE, Madrid (in Spanish) <<http://www.idae.es>>.
- Morthorst, P.E., 2001. Interactions of a tradable green certificate market with a tradable permits market. Energy Policy 29, 345–353.
- Morthorst, P.E., 2003. National environmental targets and international emission reduction instruments. Energy Policy 31, 73–83.
- Pérez Arriaga, J.A., 2005. Libro Blanco sobre la reforma del marco regulatorio de la generación eléctrica en España. Instituto de Investigación Tecnológica. Universidad Pontificia Comillas de Madrid.
- Ragwitz, M. et al., 2006. Monitoring and evaluation of policy instruments to support renewable electricity in EU Member States (final report). Energy Economics Group. Fraunhofer Institute, Germany.
- Rathmann, M., 2007. Do support systems for RES-E reduce EU-ETS-driven electricity prices? Energy Policy 35 (1), 342–349.
- Del Río, P., 2007. The interaction between emissions trading and renewable electricity support schemes: an overview of the literature. Mitigation and Adaptation Strategies for Global Change 12 (8), 1363–1390.
- del Río, P., Gual, M.A., 2004. The promotion of green electricity in Europe: present and future. European Environment Journal 14, 219–234.
- Sijm, J., 2003. Interaction of the EU Emissions Trading Directive with climate policy instruments in the Netherlands. Policy Brief INTERACT project. ECN, Amsterdam, July 2003.
- Skytte, K., 2006. Interplay between environmental regulation and power markets. European University Institute Working Papers no 2006/04, San Domenico, Italy.
- Sorrell, S., 2003. Back to the Drawing Board? Implications of the EU Emissions Trading Directive for UK Climate Policy. Executive summary, EU-funded project Interaction in EU Climate Policy (INTERACT). University of Sussex at Brighton, UK.
- The Economist, 2007. Energy Policy. A flurry of green intentions. 11 August 2007.
- Unger, T., Ahlgren, E., 2005. Impacts of a common green certificate market on electricity and CO₂ emission markets in the Nordic countries. Energy Policy 33, 2152–2163.
- Wood Mackenzie, 2006. The Impact of a Federal Renewable Portfolio Standard in the US. Washington.